

Review retrospective:
*the new germbij content — the inverse half and the forward
programme*

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Abstract

A fidelity-focused Tide-review of the eleven files that landed in the germbij arc of the **laplace** seabed on 2026-08-09/10 (PRs #93–#103): the three-file completion of the inverse half of the note’s Theorem 3.1 in the note’s own “same asymptotics” data language, and the eight landed stages of the forward-expansion programme A. The reviewer (GPT-5.6 Sol, two consults) was given the note’s TeX, the upstream Lean definitions, and the files. The *forward arc came back clean*: all eight targeted checks — exponent-split indexing, the graded-exponential truncation, the growth exponents $j+2N$ and $(N+2)(N+1)$, the unrestricted exponential remainder, the $\lambda/4$ Gaussian budget, the T_2/q form bridge, and the two truncation conventions — verified consistent, independently agreeing with the session’s own hand checks. The *inverse arc* produced four fidelity findings, none a false statement, all of the form “the Lean statement is weaker than, or the docstring broader than, the note’s claim”: the headline assumes the matched 0–2 jet (and a shared Hessian matrix) that the note derives; equality of the additive constant the note declares lost is assumed at $j = 0$; the C_c^∞ -observable premise is bridged to bare-monomial data only semantically, not by a lemma; and one module docstring claims a translated-integral identity that was dropped during its tide. All four are escalated to the human; per the review discipline (and two forward tides in flight on these files) nothing was changed — the sixteen proof-maintenance simplifications GPT suggested are recorded as follow-ups.

1 Setting

The germbij note asks when the map Φ_L , sending a loss to the asymptotic expansions of its localized posterior expectations $\langle \phi \rangle_t$ over all observables, is injective, and its Theorem 3.1 answers the nondegenerate case: Φ_L determines, and is determined by, the pair $(w^*, \{D^k L(w^*)\}_{k \geq 2})$. On 2026-08-09 a full audit of the merged corpus against the note found the recovery theorems consuming $o(q^{k-2})$ moment-*rate* hypotheses where the note’s premise is equality of expansion *families* (differences $o(t^{-\infty})$), and identified the missing pieces; the resulting programme landed as PRs #93–#95 (items B, D, C: the expansion bridge, location recovery, the analytic germ corollary — “the inverse half of Theorem 3.1 is complete”) and opened the six-stage forward programme A, of which stages 1 through 5c-i (PRs #96–#103) had landed by the time of this review. The user asked for a review of exactly this new content, focusing on fidelity of the formalised statements to the text. Eleven files at **laplace** commit **b7566bb**; this is the widest-scoped review tide so far, run as one review with the independent consult split per arc.

2 What was reviewed

Inverse arc. `Multi/ExpansionBridge.lean`: the $t = q^{-2}$ filter bridge, rate transport from superpolynomial temperature-level agreement, the homogeneity identity $\langle P \rangle_{t=q^{-2}} = q^k \cdot \text{rescaledMoment}$, and the headline theorem `smooth_jet_recovery_of_superPoly_moments`. `Multi/LocationRecovery.lean`: anchored coordinate moments vanish, located moments recover the location, and the two-loss statement `location_eq_of_superPoly_first_moments`. `Multi/AnalyticGermRecovery.lean`: power-series diagonals, the identity-theorem step `analytic_germ_eq_of_jet_eq`, and the composed corollary `analytic_germ_recovery_of_superPoly_moments`. These three carry the `\leanref` anchors on the note’s answer (i).

Forward arc. `AsymptoticPolynomial.lean` (the expansion predicate and coefficient uniqueness), `Multi/GaussianMeso.lean` (the mesoscopic window $\sqrt{q}\|z\| \leq 1$ and beyond-all-orders outer tails), `Multi/ForwardDomain.lean` (the Peano mixin and the exact exponent split), `Multi/ExpGraded.lean` (the graded expansion of the exponential correction factor), `Multi/CoeffFn.lean` (continuity and polynomial growth of the coefficient functions), `Multi/ScalarBounds.lean` (the unrestricted exponential Taylor remainder and tail bounds), `Multi/GaussAbsorb.lean` (the T_2/q form bridge and Gaussian absorption), `Multi/WindowMajorant.lean` (the q -uniform window majorant).

Informal corpus. The note’s §§1–3 (the chain, the setup defining Φ_L and the $o(t^{-\infty})$ convention, Theorem 3.1 with its proof and Corollary 3.2), the flatness proposition, the asymptotic-notation appendix; the seabed `CLAUDE.md`; the tide logs of all eleven tides; and the two archived design consults (the nondegenerate audit whose $B > D > A > C > E$ ranking scoped the arc, and the forward-programme architecture consult). The upstream definitions (`SuperPoly`, `LocalLaplaceDomain`, `HigherLaplaceDomain`, `taylorHomogeneousTerm`, `rescaledMoment`, `monomialTest`) were extracted and supplied to the reviewer alongside the files.

3 Fidelity findings

The forward arc returned **no fidelity findings**: every targeted check came back consistent, with the reviewer re-deriving the indexing ($V_{s+1} = T_{s+3}$ carrying weight $q^{s+1} = q^{k-2}$, degrees $3, \dots, N+2$ exactly), the truncation argument (the exponent polynomial has zero constant term, so outer-exponential terms $i \geq N+1$ touch only coefficients beyond N), the budget arithmetic ($-T_2 + |S| + |R| \leq -(\lambda/2 - \lambda/8 - \lambda/8)\|z\|^2 = -(\lambda/4)\|z\|^2$), and the reconciliation of the two Taylor-truncation conventions through `remConst`. The session’s independent hand checks, done before the consult returned, agreed on every point — a genuinely convergent double verification.

The inverse arc produced four findings. None says a proved statement is false; all four say the formalisation is *narrower* than the prose around it.

1. **The matched base case (largest).** The headline theorem¹ takes both losses in `HigherLaplaceDomain` packages sharing the *same* matrix H , and assumes `hbase`: equality of iterated derivatives at orders 0, 1, 2. The note’s Theorem 3.1 *derives* the Hessian from the data $(t \text{ Cov}_t[w_i, w_j] \rightarrow (H^{-1})_{ij})$. That Hessian step exists in the merged corpus in rate language (`hessian_recovery`, via the covariance limit `tendsto_normalized_covariance`) but is not composed into the `superPoly`-language headline: there is no $k = 2$ analogue of the expansion bridge for covariance data. So the theorem formalises the higher-order induction *conditional on* an already-matched 2-jet, and the “inverse half complete” gloss (and the note’s

¹`smooth_jet_recovery_of_superPoly_moments`, `ExpansionBridge.lean`:192.

answer-(i) \leanref) reads broader than this one statement warrants without its uncomposed companions.

2. **The assumed constant.** `hbase` at $j = 0$ assumes $L_1(0) = L_2(0)$ — exactly the additive constant the note proves is *lost* ($\Phi_{L+c} = \Phi_L$). A pair $L_2 = L_1 + c$ has identical normalized moment data yet is excluded by hypothesis, so the composed corollary is strictly weaker than Corollary 3.2’s inverse direction. The analytic step’s *conclusion* ($L_1y - L_10 = L_2y - L_20$ near 0) is correctly phrased modulo constants; the overreach is confined to the hypotheses, and a shift-normalization wrapper would remove it.
3. **The observable class.** The note’s premise quantifies over $\phi \in C_c^\infty(U)$; the Lean data is bare monomials and coordinates against the U -restricted Gibbs measure. Semantically a faithful weakening: the difference to any admissible $w^\alpha \chi(w)$ is supported away from the minimum and is $O(e^{-\delta t})$ by the loss gap, hence invisible beyond all orders. But that cutoff-removal implication is *not formalised*, so the headline cannot yet be applied to the note’s literal premise without an additional lemma.
4. **A docstring overclaim.** `LocationRecovery.lean`’s module docstring says the located-moment/translated-loss identity is “recorded via translation invariance of Lebesgue measure”. No such lemma is in the file; the tide log records it was dropped mid-tide (a `PiLp` instance-synthesis failure) as non-load-bearing. `locatedMoment` is a faithful *model* by construction — the reviewer confirmed the change-of-variables reading — but the docstring describes a theorem that does not exist.

Worth recording as explicit non-findings: `SuperPoly` is equivalent to the note’s $o(t^{-\infty})$ convention; the junk values of `posteriorMomentT` at $t \leq 0$ are harmless under `atTop` filters; monomial-only test data is justified by the upstream polarization theorems; the `IsSymm` hypotheses are implied by the packages’ own `ContDiff` fields (formalization debt, not a wrong assumption); and `AnalyticAt` at the minimum is correctly weaker than “analytic on U ”.

GPT-5.6 Sol: They do not yet constitute the full inverse half of Theorem 3.1 and Corollary 3.2 from the note’s literal premises, principally because: Hessian equality is assumed rather than recovered; equality of the lost additive constant is assumed; compactly supported Φ_L -data have not formally been bridged to bare monomial/coordinate moments; translated physical posterior integrals have not formally been identified with `locatedMoment`.

4 Simplifications applied

None — deliberately. The user scoped this review to fidelity, and two forward tides (the 5c-ii numerator and packaging stages) were in flight importing these files at review time; proof-body churn would have invited rebuild and merge races for zero statement-level gain. The sixteen suggestions from the two consults — all proof-maintenance: dead code, duplicated ray-path and mesoscopic-scale derivations, a posterior-linearity helper kit, a `SuperPoly`→`Tendsto` helper, internal derivation of the symmetry hypotheses, splitting the window-majorant telescope into private lemmas — are recorded in the review log for a dedicated simplification pass once programme A lands. The signature fingerprint of all eleven files is unchanged (no edits), and the build was not touched.

5 Disagreements and open questions

No Claude/GPT disagreements: the consult’s four findings coincide with the four candidate findings recorded in the review log *before* the response arrived, and its clean forward verdict matched the hand checks. What is escalated to the human is the findings themselves, with a suggested reading:

- Findings 1–3 are most naturally read not as errors but as the *remaining perimeter* of the inverse half: a $k = 2$ superPoly bridge for covariance data (composing `hessian_recovery` into the headline), a shift-normalization wrapper discharging the $j = 0$ assumption, and a cutoff-removal lemma connecting C_c^∞ data to the monomial data. Each looks like one small forward tide; together they would let the three `\leanref` anchors on answer (i) carry their weight literally.
- Finding 4 is a one-line docstring fix (or the dropped lemma, if the `PiLp` instance issue is worth revisiting).
- Until then, the honest gloss of the landed inverse half is: *given two localized nondegenerate losses with matched value, gradient and Hessian at the minimum, superpolynomially-equal monomial moment families force equal jets at all orders, and equal germs modulo the constant in the analytic case* — with location recovery as a separate, unconditional statement.

6 Lessons

The audit consult that scoped PRs #93–#95 flagged its own caveat honestly (“B+C alone would NOT justify claiming Theorem 3.1 formalised”), and the per-tide logs preserved it — but the caveat eroded between artefact layers: by the time the claim reached the completion banner in the tide log, the seabed description, and the `\leanref` tags, the conditional nature of the base case had dropped out. Reviews of “completion” tides should routinely diff the headline theorem’s *hypothesis list* against the prose claim, which is where this class of drift lives; the statement itself compiled, was proved, and is true. Second lesson: a docstring narrating a lemma that was dropped during iteration (the translated-integral form) is a specific, checkable failure mode — when a tide drops a planned lemma, the module docstring needs the same edit.

7 Follow-ups

- Three candidate forward tides closing the inverse-half perimeter: the $k = 2$ superPoly covariance bridge; the shift-normalization wrapper; the cutoff-removal lemma (C_c^∞ data $\Rightarrow$ monomial data, via the loss gap).
- The `LocationRecovery` docstring fix (or the dropped translated-integral lemma).
- The sixteen recorded proof-maintenance simplifications, as a dedicated review pass once the forward programme’s 5c-ii/packaging stages land.
- Review the forward programme’s remaining stages when they merge; this review’s forward-arc corpus (and its clean checklist) is the baseline.